

Kavach-AI Verification Node: Architecture & Overview

What is the Verification Node?

In standard AI pipelines, language models can sometimes 'hallucinate' or provide factually incorrect advice. For critical industries like Petroleum Engineering, an AI telling a worker to perform the wrong maintenance on a high-pressure pump can be catastrophic.

The Solution: The Verification Node acts as the final 'Safety Inspector' for Kavach-AI. Instead of showing the primary AI's generated report directly to the user, Kavach-AI sends the report to this Verification Node. This node is a highly-specialized, fine-tuned AI that checks the report against strict petroleum engineering rules and sensor data, intercepting any bad advice before the user ever sees it.

How We Built It

We trained a custom AI that runs entirely locally to guarantee 100% Data Sovereignty.

- 1. The Dataset:** We took thousands of rows of real-world sensor data and converted it into a specialized training format.
- 2. The Model:** We selected the lightweight but powerful Llama-3.2-3B-Instruct model.
- 3. The Training:** Using unsloth and 4-bit quantization, we fine-tuned the model directly on an RTX 4050 GPU (6GB VRAM). The model learned the exact physical limits of petroleum equipment.
- 4. The API:** We wrapped the fine-tuned model in a Lightning-fast FastAPI web server so the rest of the Kavach project can integrate with it.

How It Works in Practice

Step 1: Generation - A primary AI diagnoses a machine and generates a report.

Step 2: Verification - The system sends a POST request over the local network containing that report.

Step 3: The Verdict - The Verification Node analyzes the text. If the text is wrong, it replies with a strict REJECTED or MODIFIED correction.

Only the verified, safe output is shown on the final UI!

Why This Matters (Data Sovereignty)

Because this pipeline uses local GPUs and our own weights, no proprietary sensor data is ever sent to OpenAI, Google, or the cloud. Kavach-AI remains 100% offline, highly secure, and immune to internet outages.